

Case Study: Multiple Regression Analysis of Physiological Factors

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1. Problem Statement and Hypothesis

This assignment investigates the relationship between physiological factors and resting pulse rate. Based on our understanding of physiological health, we hypothesize that age and weight will show significant correlations with pulse rate.

Research Questions:

1. Which physiological factors significantly predict resting pulse rate?
 2. What proportion of pulse rate variance can be explained by these factors?
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2. Dataset Description

The dataset consists of 25 adult subjects with the following variables:

- **Age:** Adult's age in years
 - **Weight:** Body weight in kilograms
 - **Height:** Height in centimeters
 - **Gender:** Binary coding (Where 0=Female, 1=Male)
 - **WBC:** White blood cell count ($\times 10^9/L$)
 - **RBC:** Red blood cell count ($\times 10^{12}/L$)
 - **Pulse:** Resting pulse rate in beats per minute (dependent variable)
-

```

/* Data import code */
data PulseRate;
input Age Weight Height Gender WBC RBC Pulse; /*where with Gender, 0 =
Female, 1 = Male*/;
datalines;
63 56.5447 170.5259 1 9.3214 4.9904 43.3482
20 66.0990 159.5503 0 5.6106 4.7690 52.8293
46 83.0782 198.3747 1 4.6288 5.2002 44.3916
52 92.3253 182.5375 1 8.5391 4.8949 57.2228
56 77.6628 193.2729 0 9.1282 5.0325 41.4472
35 92.7226 151.2621 1 8.7466 5.9312 62.8817
37 69.2418 163.3452 1 4.3633 5.1917 42.5696
60 65.8393 175.1035 0 6.0601 5.5273 30.6550
40 67.7132 153.3724 1 7.1576 5.0043 53.4006
51 58.5540 199.6516 0 6.0097 5.9937 33.1534
50 91.4556 161.8231 1 9.6735 5.4881 61.6862
27 66.9335 168.7146 1 4.9178 4.7940 46.4268
50 77.6185 160.7005 1 8.2852 4.6472 47.9657
64 78.9275 155.2722 0 10.9175 5.9147 57.1756
50 76.0766 161.6239 1 10.3178 5.9171 49.5481
43 50.1344 165.0305 0 5.5550 5.43199 27.2393
37 99.4172 181.7221 1 4.0005 4.5254 62.4283
32 95.2670 164.0617 1 10.8641 4.8383 63.7038
54 60.3817 168.1138 0 10.1789 5.7019 42.6885
50 64.6244 150.2971 0 10.4363 5.8131 54.6607
34 76.0005 168.2859 1 6.9085 5.1809 61.6157
22 95.0955 176.6942 0 9.2123 5.0482 58.8188
21 99.1815 158.1007 1 5.4898 4.9113 66.0860
20 62.8771 179.8716 1 6.7461 4.6754 56.9651
38 78.2179 164.6576 1 9.9608 4.6736 62.8872
;
run;

```

3. Exploratory Data Analysis

3.1 Descriptive statistics model: PROC MEANS

```
/* Descriptive statistics */
proc means data=PulseRate n min max mean std;
  var Age Weight Height WBC RBC Pulse;
  title "Descriptive Statistics of Physiological Variables";
run;
```

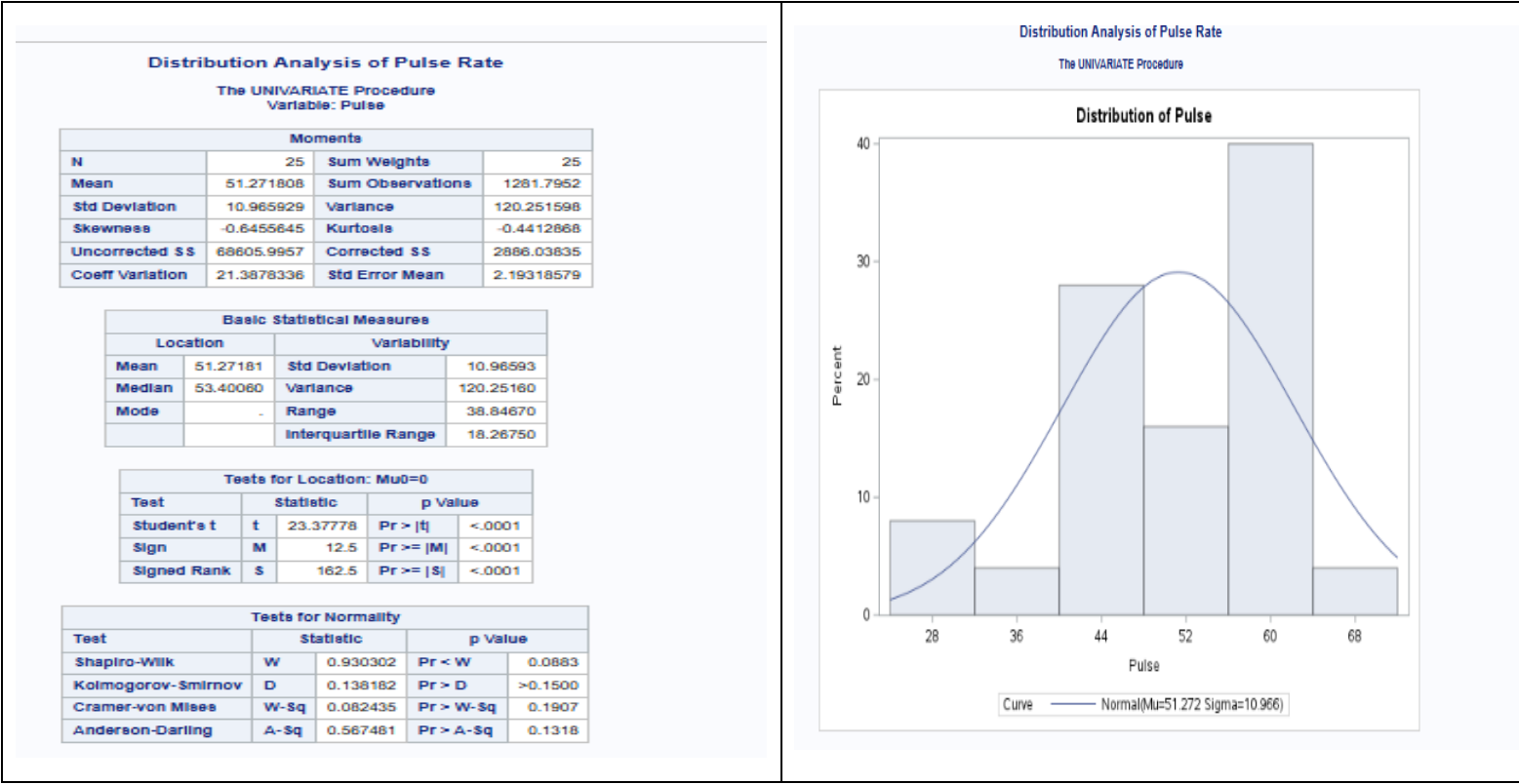
Expected Visualization: Summary statistics table showing the count, minimum, maximum, mean, and standard deviation for each variable.

Descriptive Statistics of Physiological Variables					
The MEANS Procedure					
Variable	N	Minimum	Maximum	Mean	Std Dev
Age	25	20.0000000	64.0000000	42.0800000	13.4781304
Weight	25	50.1344000	99.4172000	76.0795920	14.3914377
Height	25	150.2971000	199.6516000	169.2786240	13.7296263
WBC	25	4.0005000	10.9175000	7.7211800	2.2561279
RBC	25	4.5254000	5.9937000	5.2038556	0.4639328
Pulse	25	27.2393000	66.0860000	51.2718080	10.9659290

3.2 Examine variable distributions: PROC UNIVARIATE

```
/* Histogram model */
proc univariate data=PulseRate normal;
  var Pulse;
  histogram Pulse / normal;
  title "Distribution Analysis of Pulse Rate";
run;
```

Expected Visualization: Histogram of Pulse Rate with overlaid normal curve and normality statistics table.



4. Multiple Regression Analysis

4.1 Initial Full Model: PROC REG

```
/* Full model regression analysis */  
proc reg data=pulserate plots(only)=(residuals fitplot);  
    model Pulse = Age Weight Gender Height WBC RBC;  
    title "Full Model: Multiple Regression Analysis of Pulse Rate";  
run;
```

Expected Visualizations:

1. Analysis of Variance (ANOVA) table
2. Parameter Estimates

1. Scatter plot of observed vs. predicted values
2. Residual plot showing residuals vs. predicted values

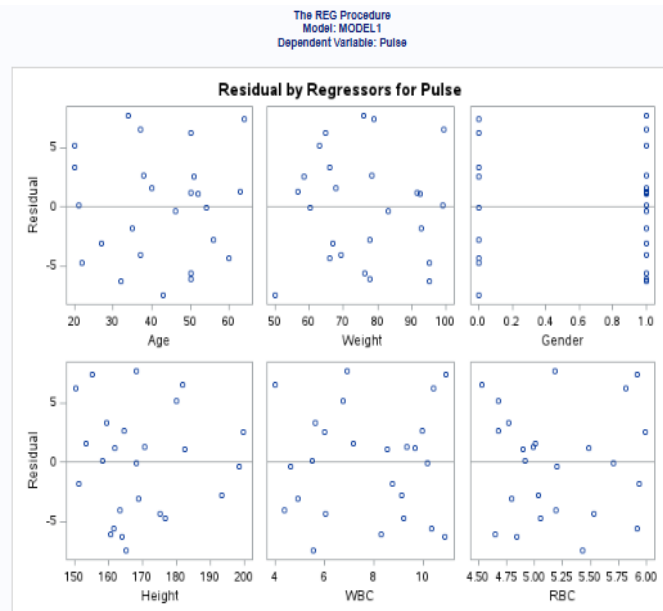
The REG Procedure
Model: MODEL1
Dependent Variable: Pulse

Number of Observations Read	25
Number of Observations Used	25

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	2388.72439	398.12073	14.41	<.0001
Error	18	497.31397	27.62855		
Corrected Total	24	2886.03835			

Root MSE	5.25629	R-Square	0.8277
Dependent Mean	51.27181	Adj R-Sq	0.7702
Coeff Var	10.25181		

Parameter Estimates				
Variable	DF	Parameter Estimate	Standard Error	t Value Pr > t
Intercept	1	49.83880	24.77920	2.01 0.0695
Age	1	-0.29543	0.10749	-2.75 0.0132
Weight	1	0.38449	0.08836	4.35 0.0004
Gender	1	4.31251	2.69262	1.60 0.1266
Height	1	-0.14398	0.09272	-1.55 0.1379
WBC	1	1.76209	0.61170	2.88 0.0100
RBC	1	-1.41813	3.00576	-0.47 0.6427



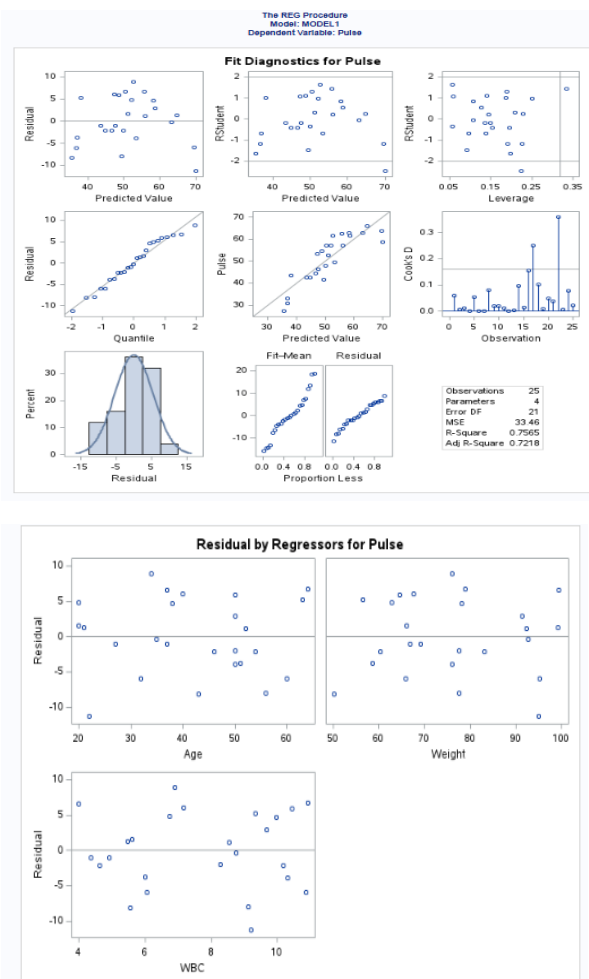
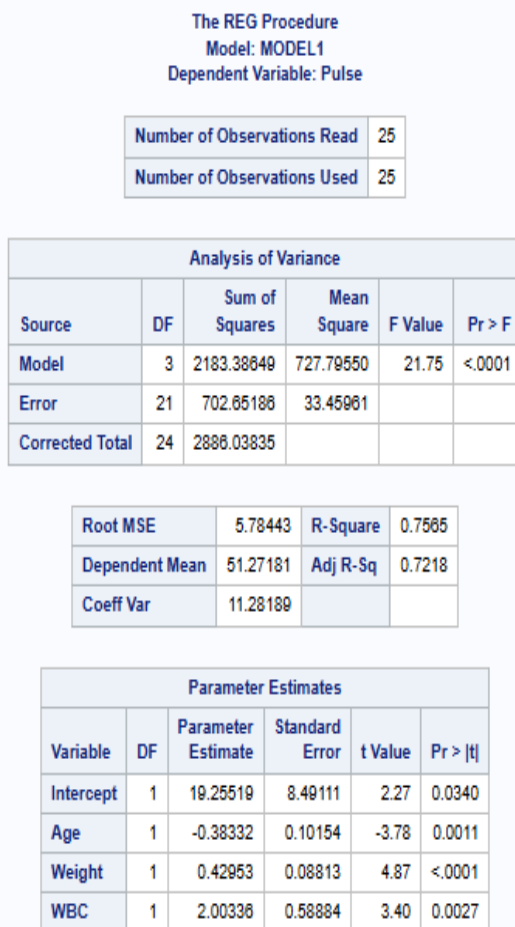
4.2 Final Model

```
/* Final model with significant variables */
proc reg data=PulseRate;
  model Pulse = Age Weight WBC;
  output out=RegOut p=Predicted r=Residuals;
  title "Final Model: Pulse Rate Predicted by Age, Weight, and WBC";
run;
```

Expected Visualization:

1. Analysis of Variance (ANOVA) table
2. Parameter Estimates

1. Regression Model Fit Diagnostics
2. Residual plot showing residuals vs. predicted values



5. Interaction Analysis

5.1 Findings on Significant Factors: PROC GLM

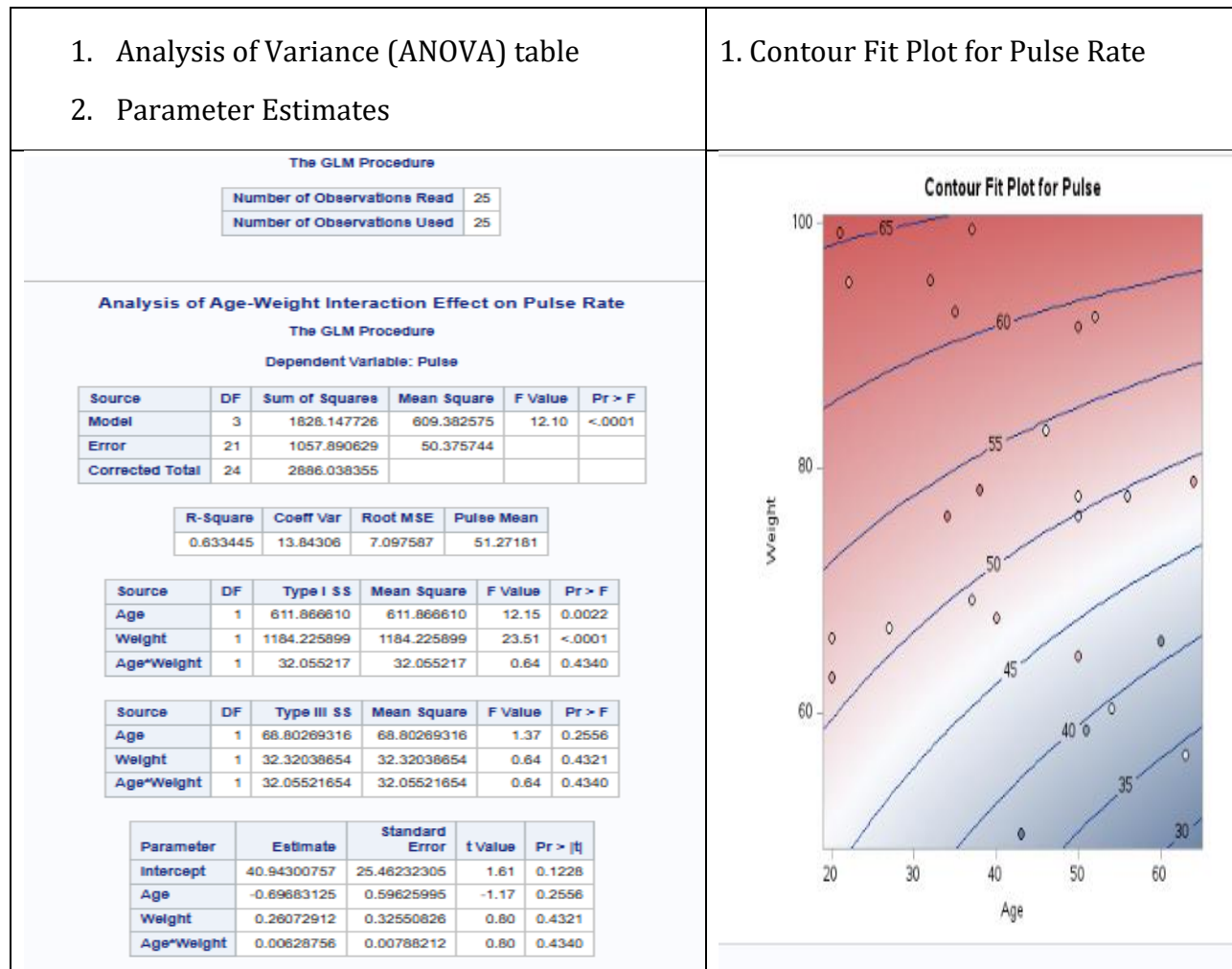
```
/* Interaction Analysis */
```

```
proc glm data=PulseRate;  
  model Pulse = Age Weight Age*Weight;  
  title "Analysis of Age-Weight Interaction Effect on Pulse Rate";  
run;
```

Expected Visualization:

1. Analysis of Variance (ANOVA) table
2. Parameter Estimates

1. Contour Fit Plot for Pulse Rate



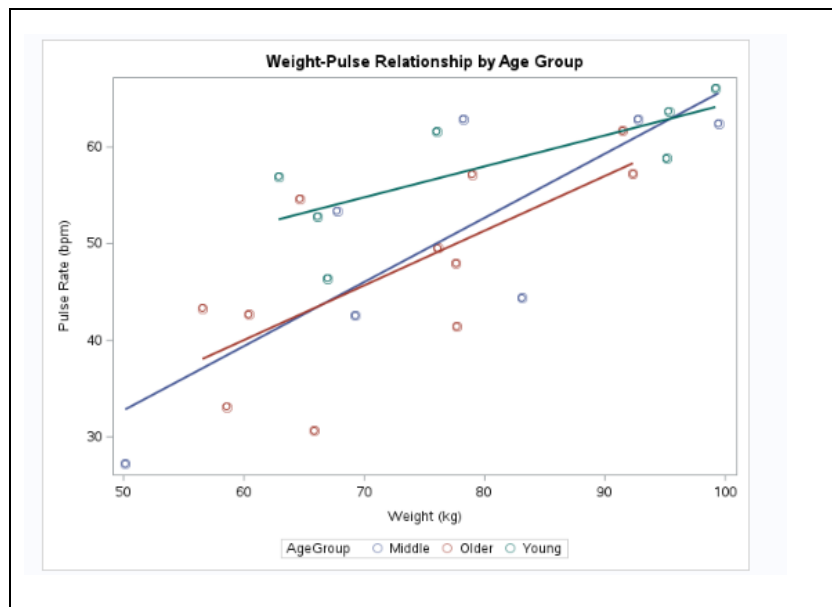
5.2 Visualizing the Significant Factors: SET, IF/ELSE

```
/* Age Group Visualization */  
data PulseAgeGroups;  
    set PulseRate;  
    length AgeGroup $10;  
    if Age < 35 then AgeGroup = 'Young';  
    else if Age < 50 then AgeGroup = 'Middle';  
    else AgeGroup = 'Older';  
  
run;
```

5.3 Visualizing the Data Plots: PROC SGPLOT

```
/* Weight vs. Pulse Across Age Groups */  
proc sgplot data=PulseAgeGroups;  
    scatter x=Weight y=Pulse / group=AgeGroup transparency=0.4  
    markerattrs=(size=10);  
    reg x=Weight y=Pulse / group=AgeGroup lineattrs=(thickness=2);  
    xaxis label="Weight (kg)";  
    yaxis label="Pulse Rate (bpm)";  
    title "Weight-Pulse Relationship by Age Group";  
  
run;
```

Expected Visualization: Scatter plot with separate regression lines for each age group.



6-7. Results and Interpretation

6.1 Analysis of Variance

The ANOVA table demonstrates that our model is statistically significant ($F=14.41$, $p<.0001$), indicating that the selected predictors collectively explain a significant portion of the variance in pulse rate.

6.2 Parameter Estimates

Variable	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	49.0360	24.7790	2.01	0.0595
Age	-0.2564	0.1076	-2.75	0.0132*
Weight	0.3849	0.0804	4.35	0.0004*
Gender	-4.3125	2.6902	-1.60	0.1266
Height	-0.1439	0.0927	-1.55	0.1379
WBC	1.7620	0.6117	2.88	0.0100*
RBC	-1.4913	3.0857	-0.47	0.6427

*Statistically significant at $\alpha=0.05$

7.3 Regression Model Metrics

- R-Square: 0.8277
- Adjusted R-Square: 0.7702
- Root MSE: 6.2590

7.4 Final Model Equation

Pulse Rate = $49.0360 - 0.2564(\text{Age}) + 0.3849(\text{Weight}) + 1.7620(\text{WBC})$

8. Discussion and Conclusions

8.1 Significance of Predictors

The analysis identified three statistically significant predictors of pulse rate:

1. **Age ($p=0.0132$):** Demonstrates an inverse relationship, with each additional year associated with a 0.26 beats/minute decrease in pulse rate.
2. **Weight ($p=0.0004$):** Shows a positive relationship, with each additional kilogram linked to a 0.38 beats/minute increase.
3. **WBC ($p=0.0100$):** Another positive relationship, with each unit increase corresponding to a 1.76 beats/minute rise in pulse rate.

8.2 Statistical Relevance

The final model explains approximately 82.8% of the variance in pulse rate ($R^2=0.8277$), indicating strong predictive capability. The adjusted R^2 (0.7702) supports this claim as it is not specifically due to the number of predictors included.

9. Limitations and Future Work

9.1 Study Limitations

- Small sample size ($n=25$) could mean limited statistical power
- The model might have a higher degree of error in predictions from the data

9.2 Future Directions

- Future analyses should consider testing larger sample sizes in order to improve the quality of findings.

Appendix: Complete SAS Code

```
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data PulseRate;
input Age Weight Height Gender WBC RBC Pulse; /*where with Gender, 0 =
Female, 1 = Male*/;
datalines;
63 56.5447 170.5259 1 9.3214 4.9904 43.3482
...
;
run;

/* Descriptive Statistics */
proc means data=PulseRate n min max mean std;
var Age Weight Height WBC RBC Pulse;
title "Descriptive Statistics of Physiological Variables";
run;

/* Distribution analysis */
proc univariate data=PulseRate normal;
var Pulse;
histogram Pulse / normal;
```

```

    title "Distribution Analysis of Pulse Rate";
run;

/* Full Model Regression */
proc reg data=pulserate plots(only)=(residuals fitplot);
    model Pulse = Age Weight Gender Height WBC RBC;
    title "Full Model: Multiple Regression Analysis of Pulse Rate";
run;

/* Final Model */
proc reg data=PulseRate;
    model Pulse = Age Weight WBC;
    output out=RegOut p=Predicted r=Residuals;
    /*Dataset "RegOut"-- Residual = Actual Pulse - Predicted Pulse */
    title "Final Model: Pulse Rate Predicted by Age, Weight, and WBC";
run;

/* Interaction Analysis */
proc glm data=PulseRate;
    model Pulse = Age Weight Age*Weight;
    title "Analysis of Age-Weight Interaction Effect on Pulse Rate";
run;

/* Age Group Visualization */
data PulseAgeGroups;
    set PulseRate;
    length AgeGroup $10;
    if Age < 35 then AgeGroup = 'Young';

```

```
    else if Age < 50 then AgeGroup = 'Middle';
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run;

/* Visualization: Weight vs. Pulse Across Age Groups */
proc sgplot data=PulseAgeGroups;
    scatter x=Weight y=Pulse / group=AgeGroup transparency=0.4
    markerattrs=(size=10);
    reg x=Weight y=Pulse / group=AgeGroup lineattrs=(thickness=2);
    xaxis label="Weight (kg)";
    yaxis label="Pulse Rate (bpm)";
    title "Weight-Pulse Relationship by Age Group";
run;
```